Software Analysis

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2026-08-31

```
knitr::opts_chunk$set(echo = TRUE)

library(here) #here() loads data
```

```
## here() starts at /Users/crispinjordan/Documents/TeachingFellowshipBioMed/PTAS_application2021/PTAS_An
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.1     v stringr   1.5.2
## v ggplot2   4.0.0     v tibble    3.3.0
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0
```

```
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(ggplot2)
library(ggtext)
library(janitor) # for clean_names()
```

```
##
## Attaching package: 'janitor'
##
## The following objects are masked from 'package:stats':
##
##     chisq.test, fisher.test
```

```
library(gridExtra) #grid.arrange() displays multiple plots
```

```
##
## Attaching package: 'gridExtra'
##
## The following object is masked from 'package:dplyr':
##
##     combine
```

```
library(scales) #label_wrap() displays axis labels over multiple lines
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
```

```
##
##     discard
##
## The following object is masked from 'package:readr':
##
##     col_factor
```

```
library(brms)
```

```
## Loading required package: Rcpp
## Loading 'brms' package (version 2.23.0). Useful instructions
## can be found by typing help('brms'). A more detailed introduction
## to the package is available through vignette('brms_overview').
##
## Attaching package: 'brms'
##
## The following object is masked from 'package:stats':
##
##     ar
```

```
library(broom.mixed) #Used in prior sensitivity analysis
library(ggrepel)  #Used in prior sensitivity analysis

set.seed=1234
```

Load **survey_data**

- survey responses from qualtrics
- includes: 6a) What format of statistical software do students learn to use?

```
#source(here("code", "cleaning_survey_data.R"))
survey_data <- read_rds(here("data/clean_data", "survey_data.rds"))

#look at data
survey_data %>%
  select(university_id, discipline, software_format) %>%
  glimpse()
```

```
## Rows: 103
## Columns: 3
## $ university_id   <fct> u90, u90, u67, u67, u66, u67, u97, u86, u93, u12, u87,~
## $ discipline      <fct> Bio, Psych, Bio, Med, Bio, Psych, Biomed, Biomed, Med,~
## $ software_format <ord> script, script, script, script, both, script, both, bo~
```

```
#structure: ordered factor
str(survey_data$software_format)
```

```
##  Ord.factor w/ 4 levels "script"<"both"<..: 1 1 1 1 2 1 2 2 1 3 ...
```

```
#table: software format by discipline
table(survey_data$discipline, survey_data$software_format)
```

```
##
##          script both push-button neither
##   Bio        17   11           3       0
##   Biomed      2    7          10       0
##   Med         5    0           4       6
##   Psych       8    4          26       0
```

```r
#filter data: remove responses where no software is taught
survey_filt <- survey_data %>%
  filter(!software_format == "neither")

#table: software format by discipline after removing responses where no software is taught
table(survey_filt$discipline, survey_filt$software_format)
```

```
##
##          script both push-button neither
##   Bio        17   11           3       0
##   Biomed      2    7          10       0
##   Med         5    0           4       0
##   Psych       8    4          26       0
```

Multinomial model of Software data

We now fit a multinomial model, using a prior recommended by Gelman et al. (2008) (DOI: 10.1214/08-AOAS191).

Below, we (1) identify model components that require a prior, (2) specify the prior, (3) specify contrasts (as recommended by Gelman el al. (2008)), (4) perform a prior predictive check.

```r
#We obtain the default prior, and then update the prior, below.
get_prior(software_format ~ discipline, family = categorical(link = "logit"), data = survey_filt)
```

```
##                 prior     class             coef group resp         dpar nlpar
##                (flat)         b                                    muboth
##                (flat)         b disciplineBiomed                   muboth
##                (flat)         b    disciplineMed                   muboth
##                (flat)         b  disciplinePsych                   muboth
##  student_t(3, 0, 2.5) Intercept                                    muboth
##                (flat)         b                                mupushbutton
##                (flat)         b disciplineBiomed               mupushbutton
##                (flat)         b    disciplineMed               mupushbutton
##                (flat)         b  disciplinePsych               mupushbutton
##  student_t(3, 0, 2.5) Intercept                                mupushbutton
##  lb ub tag       source
##                 default
##            (vectorized)
##            (vectorized)
##            (vectorized)
##                 default
##                 default
##            (vectorized)
##            (vectorized)
##            (vectorized)
##                 default
```

```r
#Our prior, following Gelman et al. (2008):
priors.software <- c(
  prior(cauchy(0, 10), class = "Intercept", dpar = "muboth"),
  prior(cauchy(0, 10), class = "Intercept", dpar = "mupushbutton"),
  prior(cauchy(0, 2.5), class = "b", dpar = "muboth"),
  prior(cauchy(0, 2.5), class = "b", dpar = "mupushbutton")
)
```

```r
contrasts(survey_filt$discipline) =contr.sum(4)
contrasts(survey_filt$discipline)
```

```
##        [,1] [,2] [,3]
## Bio       1    0    0
## Biomed    0    1    0
## Med       0    0    1
## Psych    -1   -1   -1
```

```r
#       [,1] [,2] [,3]
#Bio       1    0    0
#Biomed    0    1    0
#Med       0    0    1
#Psych    -1   -1   -1

#Prior predictive check:
brm.formula1.soft <- brmsformula(software_format ~ discipline,
                     family = categorical(link = "logit"))

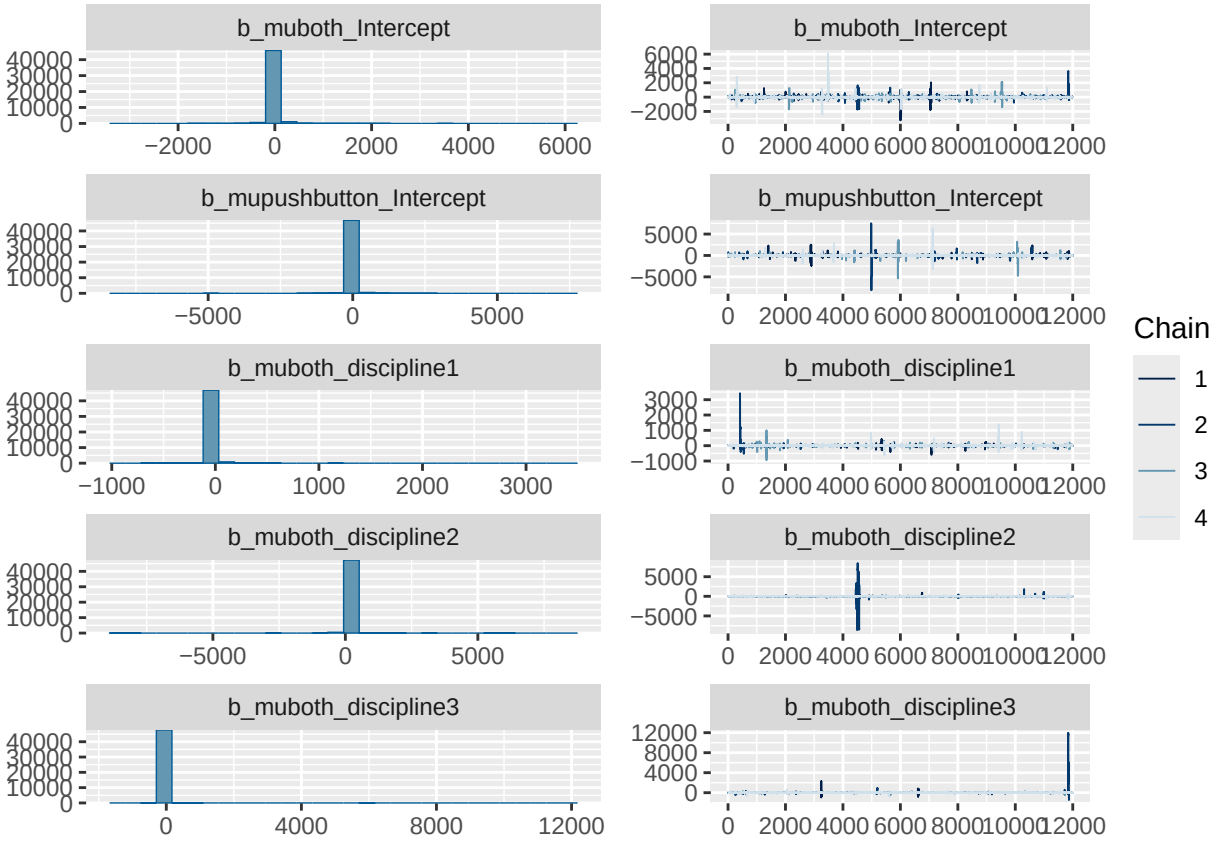
brm.mn.software.priorpc <- brm(brm.formula1.soft, prior = priors.software,
                               sample_prior = "only",
                               iter = 24000,
                     control = list(adapt_delta = 0.999, stepsize = 0.001,
                                    max_treedepth = 20),
                     data = survey_filt,
                     file = here("models.gelman", "brm.mn.software.priorpc"))

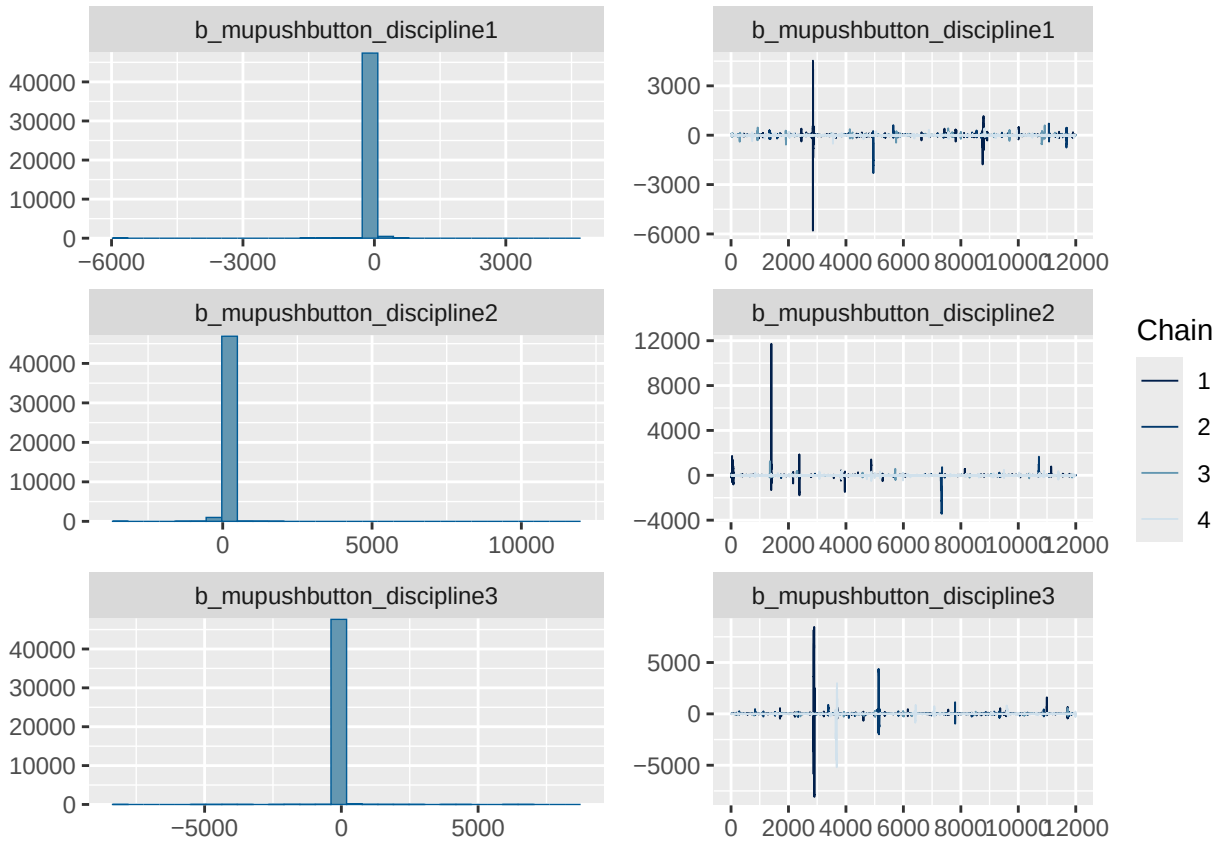
summary(brm.mn.software.priorpc)
```

```
##  Family: categorical
##   Links: muboth = logit; mupushbutton = logit
## Formula: software_format ~ discipline
##    Data: survey_filt (Number of observations: 97)
##   Draws: 4 chains, each with iter = 24000; warmup = 12000; thin = 1;
##          total post-warmup draws = 48000
##
## Regression Coefficients:
##                           Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## muboth_Intercept              3.51    179.52  -135.02   153.52 1.00    12078
## mupushbutton_Intercept        2.51    210.67  -127.58   146.45 1.00    14211
## muboth_discipline1            0.50     47.29   -29.59    27.64 1.00    27163
## muboth_discipline2            0.19    284.44   -32.73    30.93 1.00    14062
## muboth_discipline3            4.65    219.00   -29.55    30.00 1.00    19624
## mupushbutton_discipline1     -0.40     87.98   -31.70    33.67 1.00    33525
## mupushbutton_discipline2     -0.45    101.99   -28.43    29.16 1.00    19885
## mupushbutton_discipline3      1.00    223.17   -31.31    29.41 1.00    20725
##                           Tail_ESS
## muboth_Intercept              4176
## mupushbutton_Intercept        4796
## muboth_discipline1            6881
## muboth_discipline2            3999
## muboth_discipline3            5919
## mupushbutton_discipline1      7492
## mupushbutton_discipline2      5629
```

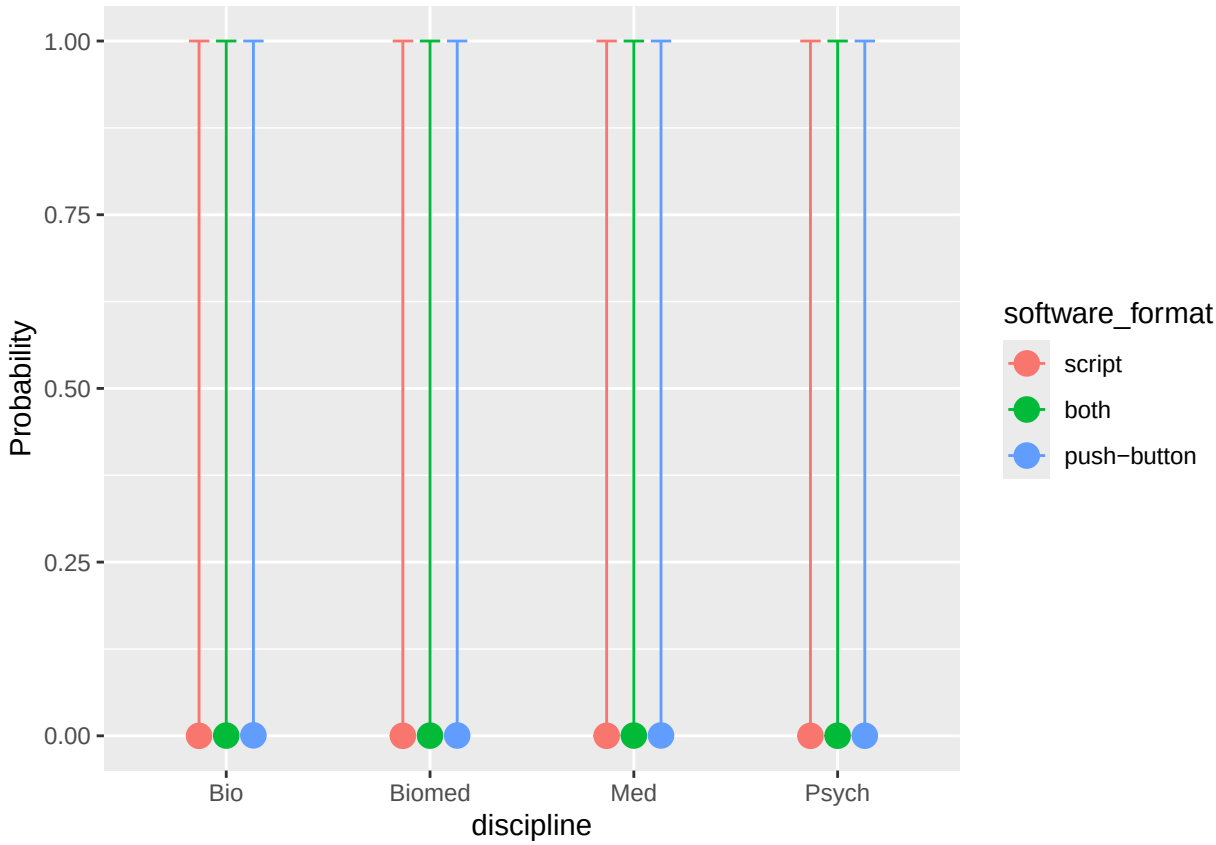
```
## mupushbutton_discipline3     5520
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
plot(brm.mn.software.priorpc)
```






```
#Prior predictive check outout:
conditional_effects(brm.mn.software.priorpc, categorical = TRUE)
```



Output from `summary(brm.mn.software.priorpc)` and `plot(brm.mn.software.priorpc)` both indicate that estimates are centered on zero (the prior has little influence on estimates).

NOTE: exploratory analyses (not shown) indicate that stronger priors yield estimates that approach 33% for each discipline (with narrower credible intervals) in prior predictive checks (`conditional_effects(brm.mn.software.priorpc categorical = TRUE)`).

Now we run the multinomial model with our specified prior, and check for appropriate model behaviour:

```
#Run the model with the Gelman prior:
brm.mn.software <- brm(software_format ~ discipline,
                       family = categorical(link = "logit"),
                       prior = priors.software,
                       iter = 6000,
                       control = list(adapt_delta = 0.999, stepsize = 0.001,
                                      max_treedepth = 20),
                       data = survey_filt,
                       file = here("models.gelman", "brm.mn.software"))

#Examine prior that we specified:
prior_summary(brm.mn.software)
```

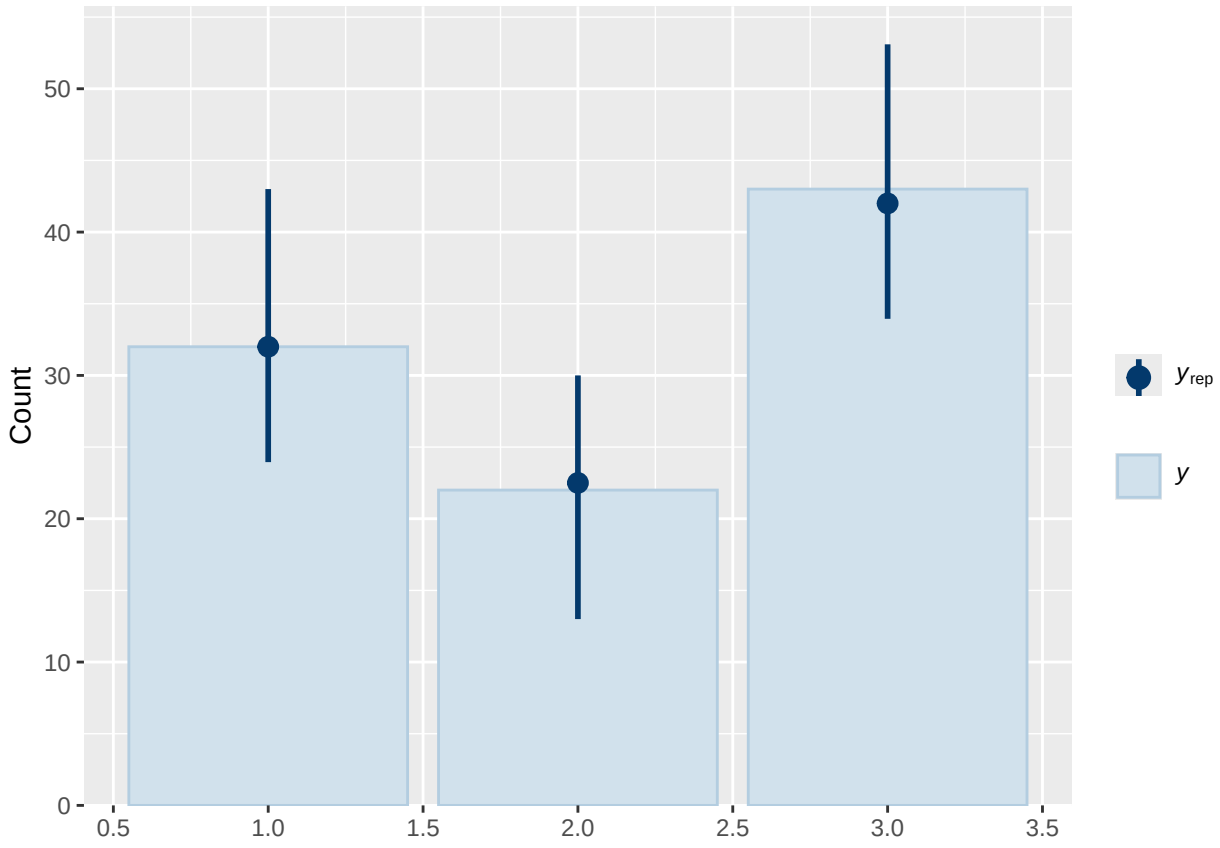
```
##           prior     class        coef group resp         dpar nlpar lb ub tag
##  cauchy(0, 2.5)         b                              muboth
##  cauchy(0, 2.5)         b discipline1                  muboth
##  cauchy(0, 2.5)         b discipline2                  muboth
##  cauchy(0, 2.5)         b discipline3                  muboth
##  cauchy(0, 2.5)         b                          mupushbutton
##  cauchy(0, 2.5)         b discipline1              mupushbutton
```

```
##  cauchy(0, 2.5)         b discipline2            mupushbutton
##  cauchy(0, 2.5)         b discipline3            mupushbutton
##   cauchy(0, 10) Intercept                           muboth
##   cauchy(0, 10) Intercept                        mupushbutton
##        source
##          user
##  (vectorized)
##  (vectorized)
##  (vectorized)
##          user
##  (vectorized)
##  (vectorized)
##  (vectorized)
##          user
##          user
```

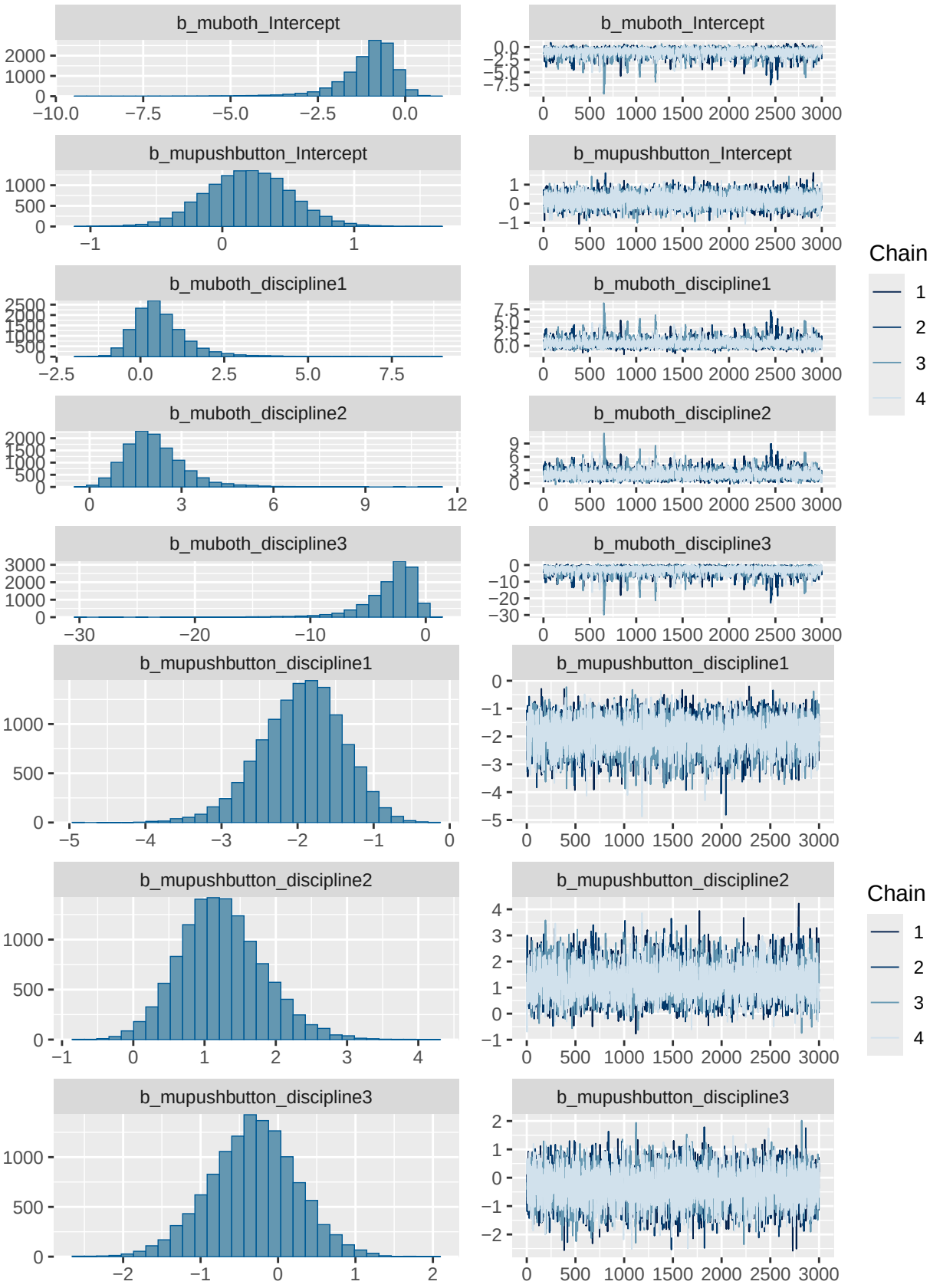
```
#Check convergence and coverage:
#We want Rhat to be less than 1.01
#We want _ESS to be 100 times the number of chains we run
summary(brm.mn.software)
```

```
##  Family: categorical
##   Links: muboth = logit; mupushbutton = logit
## Formula: software_format ~ discipline
##    Data: survey_filt (Number of observations: 97)
##   Draws: 4 chains, each with iter = 6000; warmup = 3000; thin = 1;
##          total post-warmup draws = 12000
##
## Regression Coefficients:
##                          Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
## muboth_Intercept            -1.07      0.86    -3.27     0.04 1.00     2382
## mupushbutton_Intercept       0.19      0.32    -0.43     0.82 1.00     6988
## muboth_discipline1           0.62      0.89    -0.61     2.85 1.00     2497
## muboth_discipline2           2.12      1.02     0.61     4.56 1.00     2388
## muboth_discipline3          -3.08      2.46    -9.50    -0.29 1.00     2319
## mupushbutton_discipline1    -1.95      0.54    -3.09    -0.97 1.00     5782
## mupushbutton_discipline2     1.24      0.59     0.17     2.49 1.00     5399
## mupushbutton_discipline3    -0.32      0.57    -1.46     0.78 1.00     6445
##                          Tail_ESS
## muboth_Intercept             1520
## mupushbutton_Intercept       7323
## muboth_discipline1           1523
## muboth_discipline2           1696
## muboth_discipline3           1418
## mupushbutton_discipline1     6155
## mupushbutton_discipline2     6142
## mupushbutton_discipline3     7097
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
#Posterior predictive check:
pp_check(brm.mn.software, type = "bars", ndraws = 100)
```




```
#Check traces
plot(brm.mn.software)
```



The summary (`summary(brm.mn.software)`) indicates that the model converged well (based on Rhat) and

has high coverage for all estimates.

The posterior predictive check output (`pp_check(brm.mn.software, type = "bars", ndraws = 100)`) indicates sufficient model fit.

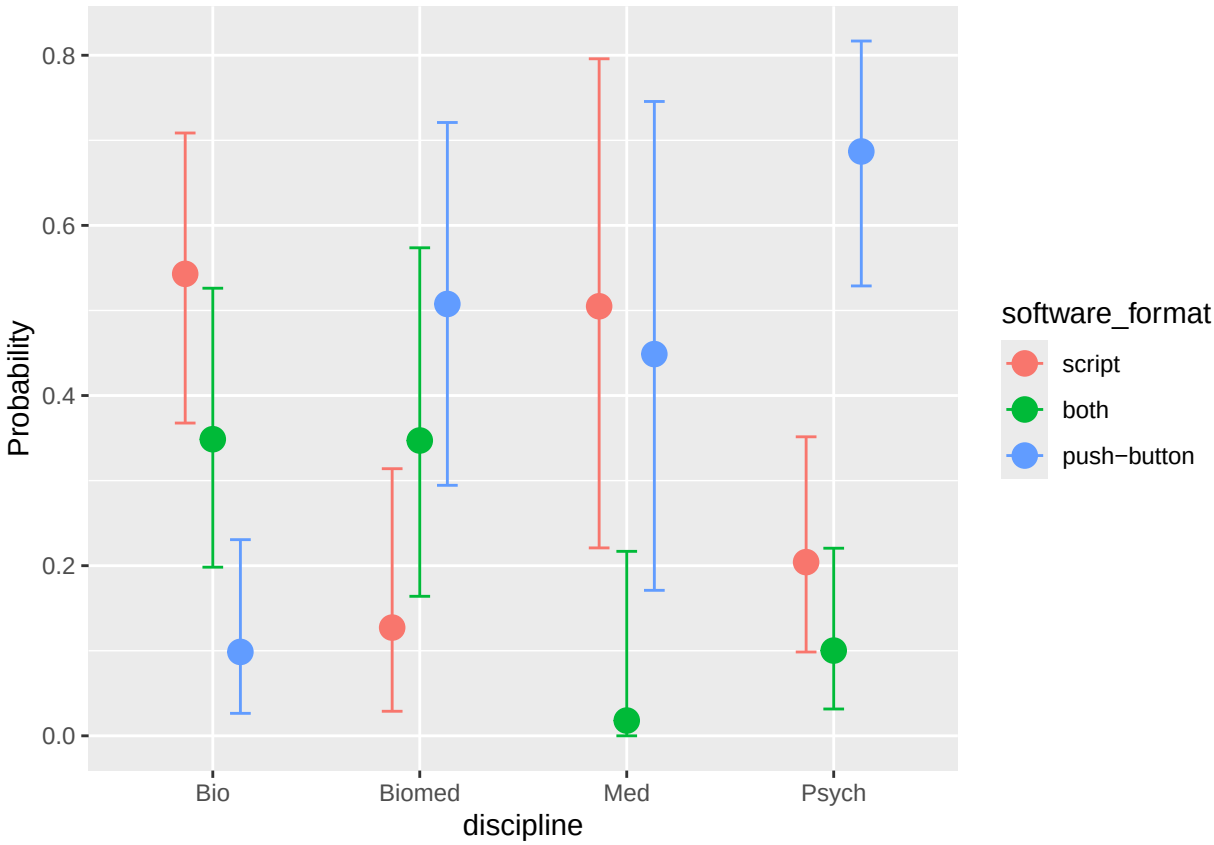
The traces and distributions of estimates on the latent scale (`plot(brm.mn.software)`) also indicate good model behaviour.

Now we obtain the model estimates:

```
eff_brm.mn.software <- conditional_effects(brm.mn.software, categorical = TRUE)
eff_brm.mn.software[[1]]
```

```
##    discipline software_format cond__      cats__ effect1__   effect2__
## 1         Bio          script      1      script       Bio      script
## 2      Biomed          script      1      script    Biomed      script
## 3         Med          script      1      script       Med      script
## 4       Psych          script      1      script     Psych      script
## 5         Bio          script      1        both       Bio        both
## 6      Biomed          script      1        both    Biomed        both
## 7         Med          script      1        both       Med        both
## 8       Psych          script      1        both     Psych        both
## 9         Bio          script      1 push-button       Bio push-button
## 10     Biomed          script      1 push-button    Biomed push-button
## 11        Med          script      1 push-button       Med push-button
## 12      Psych          script      1 push-button     Psych push-button
##    estimate__       se__      lower__   upper__
## 1  0.54302285 0.08984112 3.676498e-01 0.7086097
## 2  0.12724243 0.07140911 2.886793e-02 0.3140023
## 3  0.50493111 0.15924970 2.209022e-01 0.7958318
## 4  0.20418575 0.06326488 9.854228e-02 0.3515034
## 5  0.34864595 0.08639394 1.982455e-01 0.5261468
## 6  0.34714001 0.10898941 1.640086e-01 0.5736769
## 7  0.01795076 0.02583342 1.450166e-06 0.2168489
## 8  0.10022473 0.04810416 3.153521e-02 0.2205281
## 9  0.09863369 0.05192203 2.641674e-02 0.2305951
## 10 0.50764370 0.11145234 2.944635e-01 0.7209700
## 11 0.44872766 0.16127035 1.710362e-01 0.7456501
## 12 0.68690557 0.07313738 5.288563e-01 0.8167323
```

```
print(eff_brm.mn.software)
```



Modify data.frame

You may note that, above, the output of `table(survey_filt$discipline, survey_filt$software_format)` included a column for `Neither`, even though this column contained no observations. As a result, if we use `survey_filt` (in its current form) to plot the data, the plot includes a label for `neither`. Below, we modify `survey_filt` to ensure that a plot only includes categories of data that were included in the analysis (i.e., excludes `neither`).

We will do this by recoding the categories. We arbitrarily pool `neither` with `push-button`, which results in no change in the actual counts because `neither` contains no observations (we removed them, above).

First, let's look again at the original version of `survey_filt` to allow easy comparison with our change:

```
table(survey_filt$discipline, survey_filt$software_format)
```

```
##
##          script both push-button neither
##   Bio        17   11           3       0
##   Biomed      2    7          10       0
##   Med         5    0           4       0
##   Psych       8    4          26       0
```

We see the data are preserved, but the empty `neither` category dropped.

```
#recode to remove 'neither' category
survey_filt <- survey_filt %>%
  mutate(software_format_3 =
           fct_recode(software_format,
            "script" =  "script",
```

```
            "both" = "both",
            "push-button" = "push-button",
            "push-button" = "neither"
            )
         )

#table of teaching approach (3 categories) by discipline
table(survey_filt$discipline, survey_filt$software_format_3)
```

```
##
##           script both push-button
##   Bio         17   11           3
##   Biomed       2    7          10
##   Med          5    0           4
##   Psych        8    4          26
```

Plot

```
plot_colour <- "skyblue4"

plot_theme <- theme_bw() +
  theme(
    plot.title = element_text(size = 18),
    axis.title.x = element_text(size = 18),
    axis.text.x = element_text(size= 16),
    axis.title.y = element_text(size= 24),
    axis.text.y = element_markdown(size = 16, hjust = 1),
    axis.ticks.y = element_blank(),
    strip.text = element_text(size = 20),
    strip.background = element_blank(),
    panel.border = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    legend.position = "none")
```

```
software_estimates <- eff_brm.mn.software[[1]] %>%
  clean_names() %>%
  rename(
    soft_format = cats,
    probability_estimate = estimate) %>%
  select(discipline, soft_format, probability_estimate, lower, upper)

software_estimates
```

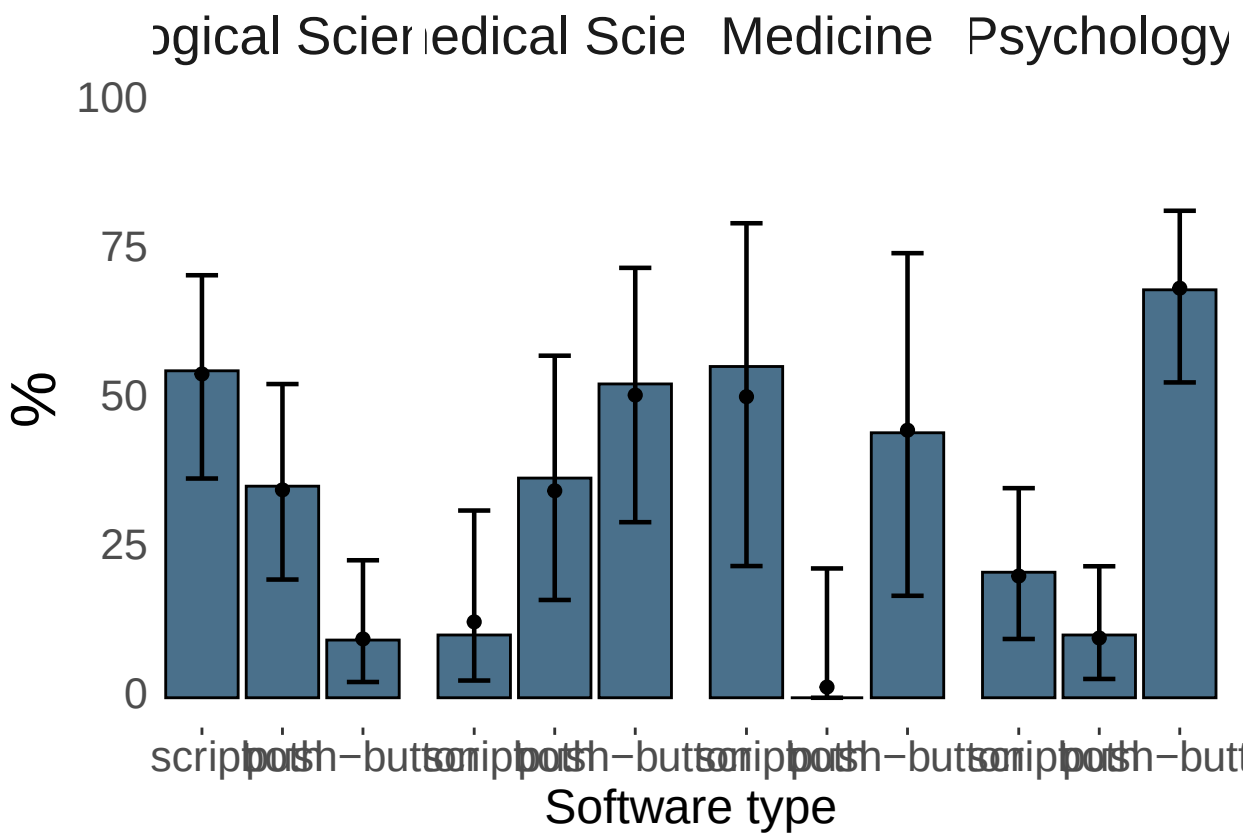
```
##    discipline soft_format probability_estimate        lower     upper
## 1         Bio      script           0.54302285 3.676498e-01 0.7086097
## 2      Biomed      script           0.12724243 2.886793e-02 0.3140023
## 3         Med      script           0.50493111 2.209022e-01 0.7958318
## 4       Psych      script           0.20418575 9.854228e-02 0.3515034
## 5         Bio        both           0.34864595 1.982455e-01 0.5261468
## 6      Biomed        both           0.34714001 1.640086e-01 0.5736769
## 7         Med        both           0.01795076 1.450166e-06 0.2168489
## 8       Psych        both           0.10022473 3.153521e-02 0.2205281
```

```
## 9         Bio push-button           0.09863369 2.641674e-02 0.2305951
## 10     Biomed push-button           0.50764370 2.944635e-01 0.7209700
## 11        Med push-button           0.44872766 1.710362e-01 0.7456501
## 12      Psych push-button           0.68690557 5.288563e-01 0.8167323
```

```
survey_table_soft <- table(survey_filt$software_format_3, survey_filt$discipline)
survey_table_soft <- prop.table(survey_table_soft, margin = 2)
survey_table_soft <- as.data.frame(survey_table_soft)
survey_ggplot_soft <- survey_table_soft %>%
  rename(
    soft_format = Var1,
    discipline = Var2) %>%
  mutate(percentage = 100*Freq)
```

```
software_plot <- ggplot() +
  geom_col(
    data = survey_ggplot_soft,
    aes(x = soft_format, y = percentage),
    color = "black",
    fill = plot_colour) +
  geom_errorbar(
    data = software_estimates,
    aes(x = soft_format, ymin = 100*lower, ymax = 100*upper),
    width = .4,
    linewidth = .8) +
  geom_point(
    data = software_estimates,
    aes(x = soft_format, y = 100*probability_estimate),
    size = 2) +
  scale_x_discrete(drop = FALSE) +
  scale_y_continuous(
    "%",
    limits = c(0, 100),
    labels = c(0, 25, 50, 75, 100)) +
  facet_wrap(
    ~ factor(
      discipline,
      labels = c("Biological Sciences", "Biomedical Sciences", "Medicine", "Psychology")),
    nrow = 1) +
  xlab("Software type") +
  plot_theme


software_plot
```




```
ggsave(
  plot = software_plot,
  here("figures", "software_plot_gelman.jpg"),
  device = "jpeg",
  width = 1280,
  height = 637,
  units = "px",
  dpi = 100)
```

Perform Sensitivity Analysis:

```
#Sensitivity analysis:  determines the extent to which the prior affects the results
#(a qualitative test, not a quantitative test)

brm.mn_fixed <- tidy(brm.mn.software, effects = "fixed", conf.level = 0.95, fix.intercept = FALSE) %>%
  mutate(
    ci_width = abs(conf.low - conf.high)
  )
```

```
## Warning in tidy.brmsfit(brm.mn.software, effects = "fixed", conf.level = 0.95,
## : some parameter names contain underscores: term naming may be unreliable!
```

```
brm.mn_fixed
```

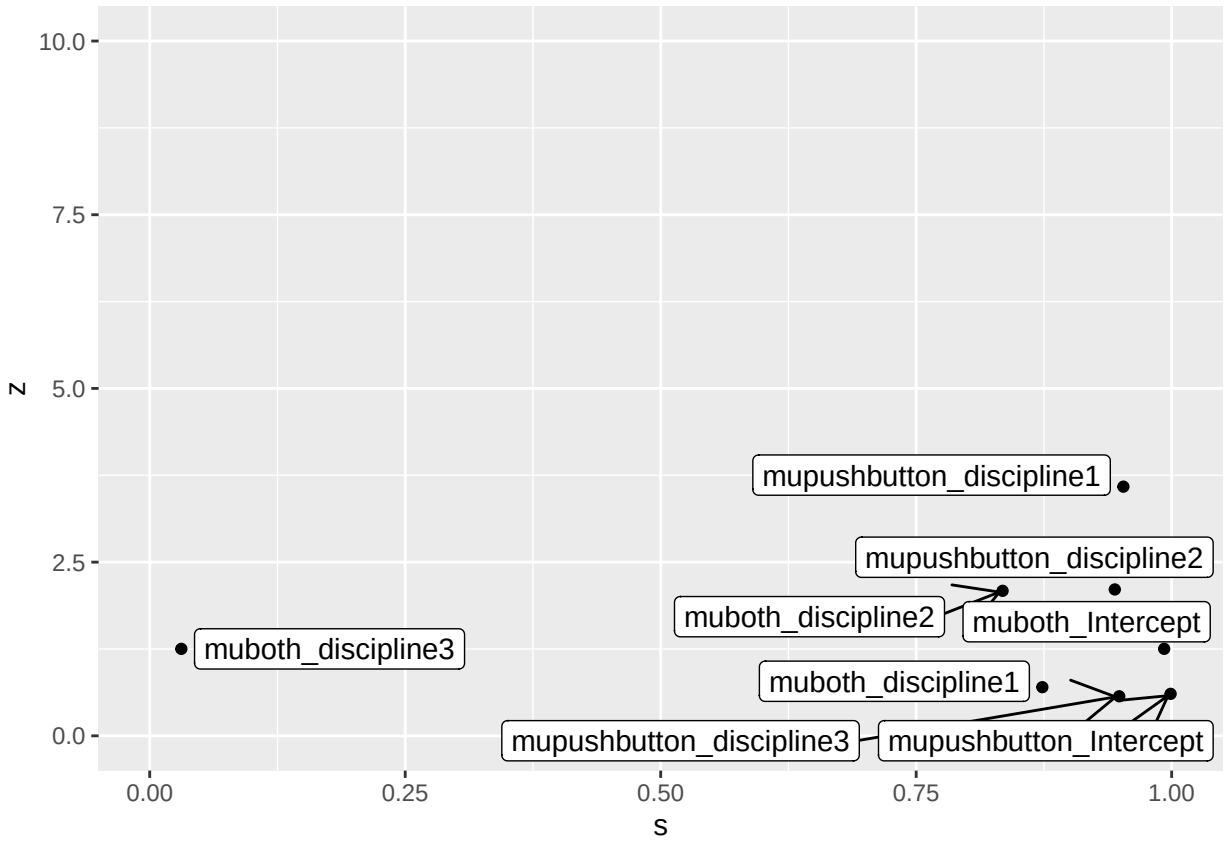
```
## # A tibble: 8 x 8
##   effect component term           estimate std.error conf.low conf.high ci_width
##   <chr>  <chr>     <chr>             <dbl>     <dbl>    <dbl>     <dbl>    <dbl>
## 1 fixed  cond      muboth_Interc~   -1.07      0.856   -3.27     0.0395     3.31
## 2 fixed  cond      mupushbutton_~    0.194     0.322   -0.429    0.824      1.25
```

```
## 3 fixed  cond      muboth_discip~    0.623     0.889   -0.612    2.85       3.46
## 4 fixed  cond      muboth_discip~    2.12      1.02     0.609    4.56       3.95
## 5 fixed  cond      muboth_discip~   -3.08      2.46    -9.50    -0.287      9.21
## 6 fixed  cond      mupushbutton_~   -1.95      0.543   -3.09    -0.967      2.12
## 7 fixed  cond      mupushbutton_~    1.24      0.590    0.167    2.49       2.32
## 8 fixed  cond      mupushbutton_~   -0.323     0.566   -1.46     0.782      2.24
```

```
#We cannot see the relevant 'term' names.  So we'll look at fewer columns:
brm.mn_fixed_reduced <- brm.mn_fixed %>%
  select(
    effect, component, term, estimate
    )
brm.mn_fixed_reduced
```

```
## # A tibble: 8 x 4
##   effect component term                     estimate
##   <chr>  <chr>     <chr>                       <dbl>
## 1 fixed  cond      muboth_Intercept           -1.07
## 2 fixed  cond      mupushbutton_Intercept      0.194
## 3 fixed  cond      muboth_discipline1          0.623
## 4 fixed  cond      muboth_discipline2          2.12
## 5 fixed  cond      muboth_discipline3         -3.08
## 6 fixed  cond      mupushbutton_discipline1   -1.95
## 7 fixed  cond      mupushbutton_discipline2    1.24
## 8 fixed  cond      mupushbutton_discipline3   -0.323
```

```
#We have 8 fixed effects; the first 2 are intercepts, and the last 6 are coefficients (b)
#Also, the 2 Intercepts are listed first in the output of brm.mn_fixed,
#and the Intercepts had priors with sd = 10
#The remaining 6 fixed effects are for b, which had sd = 2.5 for the prior.
#Hence, we set up the sencitivity analysis like this:
brm.mn_fixed %>%
  mutate(
    theta = c(0, 0, 0, 0, 0, 0, 0, 0),
    sigma_prior = c(10, 10, 2.5, 2.5, 2.5, 2.5, 2.5, 2.5),
    z = abs((estimate - theta) / std.error), # it's called here std.error but is the standard deviation
    s = 1 - (std.error^2 / sigma_prior^2)
  ) %>%
  ggplot(aes(s, z, label = term)) +
  geom_point() +
  geom_label_repel(arrow = arrow()) +
  xlim(0, 1) + ylim(0, 10)
```


10.0
7.5
5.0
2.5
0.0
z
0.00
0.25
0.50
0.75
1.00
s
mupushbutton_discipline1
mupushbutton_discipline2
muboth_discipline2
muboth_Intercept
muboth_discipline3
muboth_discipline1
mupushbutton_discipline3
mupushbutton_Intercept